AMETECS INDIA PRIVATE LIMITED

DOCUMENT 1 OF 2 · BUSINESS

Business Logic Specification

Ametecs Managed Mobile Platform — what it does, for whom, and the rules it follows

Product	Ametecs Managed Mobile Platform (working name; project "Caller" / Smart caller)
Document	Business logic — end to end
Companion	Document 2: Technical Logic & Specifications
Owner	Mohammad Ejaz Hussain, Managing Director
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1. Executive summary

The **Ametecs Managed Mobile Platform** (working name; internal project name "Caller" / "Smart caller") lets a company hand out company-owned Android phones to its field and calling staff, control how those phones are used, and run customer calling from them — with every call logged, timed, recorded and tied to a customer record. It is built first for collection and recovery agencies, the market Ametecs knows best, but works for any business that calls customers from company phones (sales, telecalling, service follow-ups, field verification).

The platform has three parts that work as one system:

Company server
The brain. Holds employees, phones, customers, calls, recordings, rules and the audit trail. Runs in the Ametecs cloud (many companies) or on the client's own server (one company).

Admin console
A web console (also packaged as a Windows desktop app) for owners, IT, supervisors and auditors. Approves phones, uploads customers, listens to recordings, chats and calls employees, watches location.

Employee app
One Android app for all client companies. The employee signs in, sees only their assigned customers, calls them from the phone's SIM, records the outcome and follow-up, and receives supervisor messages and calls.

What makes it different from a plain dialer or a plain device-management tool is that it joins three things that are normally sold separately: **(1) control of the company phone**, **(2) controlled customer calling with mandatory recording**, and **(3) field-force attendance and location** connected to SmartPRS2 (payroll) and SmartEPT (enforcement).

1.1 The problems it solves

Business problem	How the platform answers it
Agents call customers from personal phones; the company loses the number, the history and the recording when the agent leaves.	Calls are placed only from approved company phones. Every call is stored on the server with its customer, duration, outcome and recording.
Agents skip recording, or switch it off for "difficult" calls.	Recording is mandatory. The app checks that the phone's automatic call recorder is ON and blocks calling until a recording is proven to exist.
Agents dial numbers that are not in their portfolio, or call outside permitted hours.	The server decides the number from the customer record. Calls to unassigned customers, outside calling hours or to blocked prefixes are refused.
Field staff mark attendance from home.	Sign-in captures GPS. The SmartPRS2 geofence decides whether the employee is inside the allowed area; outside means sign-in is blocked (configurable).
Supervisors cannot reach agents quickly.	Built-in chat with read receipts, desktop/phone notifications, and an internet voice call from the admin console to the employee's phone.
Lost phone or employee leaves with company data.	"Revoke access" kills all sessions on the next request. Customer data on the phone disappears at the next sync.

2. Who uses it — roles and permissions

Every company gets five ready-made roles. Each role is a bundle of permissions; the company administrator can see everything. Supervisors see **only the teams they supervise** unless they are given the "see all teams" permission.

Role	Typical person	What they can do
Company Administrator	Owner / MD / operations head	Everything in the console (all permissions except using the phone app).
IT / Device Administrator	IT in-charge	Approve, revoke and wipe phones; create and publish policies; branding; storage; audit; system health; calling configuration.
Supervisor	Team leader	See own team's employees, phones, customers, calls; play recordings; assign customers; view location; chat and call employees; request screen/camera sessions; export reports.
Employee	Field agent / telecaller	Use the mobile app, make customer calls, chat.
Read-only Auditor	Compliance / client auditor	View phones, policies, employees, customers, calls, audit logs and health — cannot change anything.

There are 30 individual permissions (for example *recordings.play* and *recordings.download* are separate, so a supervisor can listen but not download). Every privileged action is written to the audit log.

3. Key terms in plain words

Term	Meaning
Company (tenant)	One client business. Its data is completely separate from every other company on the same server.
Managed device	A company phone that has registered with the server. It has an approval state: Pending, Approved, Rejected or Revoked.
Approval code	A code like K7QM-4XP2 shown on the phone. The admin types it in the console to prove they are approving the phone in front of them.
Work session	The period between employee sign-in and sign-out on an approved phone.
Policy profile	A named set of rules (calling hours, recording, location, camera, apps…) applied to the whole company, a team or one phone.
Compliant	The phone has applied and verified every mandatory rule. Only a compliant phone can show customers and make calls.
Disposition	The outcome the employee records after a call (Promise to pay, Call back later, Not reachable, Wrong number, Refused, Already paid…).
Follow-up	A dated reminder to call the customer again, created from a disposition.
Geofence	The allowed area for an employee (a point plus a radius), owned by SmartPRS2.
Offline lease	How long the employee may keep working without internet (default 4 hours). After that, work locks until the phone reconnects.
Revoke	Permanently cut a phone's access (lost phone, leaver). Does not wipe the phone.

4. The end-to-end journey

This is the full life of the system, from a new phone in a box to a recorded, dispositioned customer call that shows up on the owner's dashboard.



Behind the scenes, sign-in and sign-out are also sent to SmartPRS2 as attendance punches and to SmartEPT for monitoring, and every GPS fix during the work session is sent to SmartEPT's field trail.

5. Phone onboarding and approval

5.1 Business rules

1. The employee opens the app on a company phone and taps **Connect / Register**. The phone creates its own secret key that never leaves the phone, and the server records a **Pending** request with an approval code.
2. In **Devices & approvals** the administrator clicks *Review*, asks the employee to read the code aloud, types it, chooses the employee, team, asset tag and (optionally) a policy profile, and approves. A wrong code is refused — this stops someone approving the wrong phone.
3. The phone moves on by itself, gets a short-lived device session, and shows the sign-in screen.
4. A phone can only be used by the employee it is assigned to. Another employee signing in on it gets: "This phone is not assigned to you."

5.2 Lifecycle rules

Event	What happens
Employee signs out	Only the work session ends. The phone stays approved and its rules stay in force.
Reassign to another employee	Old employee's sessions end; the new employee must sign in.
Revoke (lost phone / leaver)	All credentials die immediately. Active calls are cut (or allowed to finish, per policy). The phone shows "Access revoked". It is not wiped.
Re-approve after revoke	Not allowed on the old registration. The phone must register again with a new key — the old key can never come back.
App reinstalled / data cleared	Appears as a brand-new pending request that needs approval again.
Wipe (factory reset)	Separate permission, typed asset-tag confirmation, a reason, audited. Only on fully managed (Android Enterprise) phones.

6. Employee sign-in, attendance and geofence

Sign-in is the moment the employee "starts work". The rules, in order:

1. User ID and password must be correct, and the account must be active and allowed to use the mobile app.
2. The phone must be approved and assigned to this employee.
3. The phone sends its current GPS position. The Caller server asks **SmartPRS2**: is this employee inside their geofence?
4. If SmartPRS2 says **outside** and the company has "Block sign-in outside the geofence" ticked, sign-in is refused with a clear message, and the attempt is audited. Nothing is recorded as attendance.
5. If SmartPRS2 cannot be reached, the employee is **allowed in** with status "unverified" (fail-open), so a server problem never locks out the whole field team.
6. On success, a work session starts and a **login punch** is queued for SmartPRS2 (attendance of record for payroll) and SmartEPT (monitoring). Sign-out creates the matching **logout punch**.

Geofence result	Meaning	Sign-in
within	Inside the allowed radius	Allowed
outside	Beyond radius (+ the rule's tolerance: strict / 1 km / 2 km)	Blocked if enforcement is on; else allowed and flagged
no-gps	Phone sent no position	Per SmartPRS2 rule
no-rule	No geofence configured for this employee	Allowed
unknown-employee	Employee code not found in SmartPRS2	Allowed; punch held in PRS "Pending Punches" for mapping
unverified	SmartPRS2 unreachable	Allowed (fail-open)

Matching rule. Employees are matched across the three products by **employee code**. The Caller User ID must equal the SmartPRS2 Emp. Code and the SmartEPT Employee Code (not case-sensitive).

7. Customer data: import, assignment and withdrawal

7.1 Import

1. Download the sample template, fill it in, upload CSV or XLSX (up to 20 MB).
2. The system recognises common column names automatically (e.g. "Loan No", "Account", "Borrower", "Mobile", "Bucket", "PTP date") and lets the admin correct the mapping.
3. *Preview*, then *Validate whole file*. Rejected rows are listed with the reason and can be downloaded as an error report.
4. *Commit* creates or updates customers. A re-import of the same customer reference updates (or skips, by choice) instead of duplicating.

Field	Rule
Customer ref	Required, unique per company (loan/account number).
Name, Phone	Required. Phone numbers are converted to international format (+91XXXXXXXXXX); Indian numbers need exactly 10 digits after +91.
Alternate phone	Optional, same rules.
Campaign / portfolio / bucket	Optional grouping.
Assigned employee / team	Optional; employee code or team code.
Status	new, in_progress, follow_up, closed, do_not_call.

Field	Rule
Follow-up date, notes	Optional.

7.2 Assignment and withdrawal

- A customer is assigned to one employee, or to a team (then anyone in that team can call).
- Supervisors can only assign within their own teams.
- Withdrawn customers disappear from the phone at its next sync; their call history is kept on the server.
- Every assignment, reassignment and withdrawal is kept as history (who, when, from which import).
- Exports are cleaned against spreadsheet formula injection.

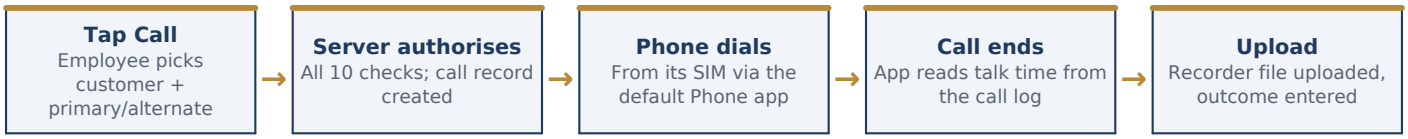
8. Calling rules

The single most important business rule: **the employee never chooses the number — the server does.** The app asks to call a customer (primary or alternate number); the server checks everything below and only then returns the number to dial.

#	Check	If it fails the employee sees
1	Calling is enabled in the policy	Calling is disabled by policy.
2	Phone is compliant with all mandatory rules	Device is not compliant with mandatory policy.
3	No other call in progress for this employee	Finish the current call first.
4	Inside calling hours (company time zone; overnight windows supported)	Outside permitted calling hours.
5	Customer is assigned to the employee (or to their team)	Customer is not assigned to you.
6	Customer status is not "do not call" or "closed"	Customer status does not permit calls.
7	Alternate number exists, if alternate chosen	No alternate number on record.
8	Any number the phone sends matches the record exactly (tamper check)	Requested number does not match the customer record.
9	Number starts with an allowed prefix (default +91)	Destination is not in the allowed prefixes.
10	Manual dialling of unassigned numbers only if the policy allows (off by default)	Manual dialling is not permitted.

8.1 How a call is placed today (SIM route)

The current, decided route is **SIM calling through the phone's own dialer**, because it needs no telephony server, works on any SIM plan, and the phone's built-in recorder captures both voices.



- **OPPO, OnePlus, Realme**: ODialer (the OPPO Phone app) must be the default Phone app. If missing, the app opens the Play Store; if not default, it opens Default Apps. Calling stays blocked until both are done.
- **Other brands** (Samsung, Xiaomi, Vivo, others): any default Phone app is accepted, as long as it auto-records calls.
- If the app is killed mid-call and never reports back, the call is closed as "result not reported" at the next call attempt, so the employee is never stuck; a late result can still complete it.

8.2 Mandatory recording gate

Android does not allow one app to read or switch another app's auto-record setting. So the platform asks for **proof**: a call-recording file must exist on the phone.

1. Every phone starts **blocked** for calling when the company requires recording.
2. The employee switches ON auto call recording in the Phone app (the app opens that settings screen for them) and makes one test call.
3. As soon as a new call-recording file appears, calling is **unlocked**.
4. If any answered call later ends without a recording, calling is **blocked again** with the message: "Recording not saved — auto call recording is OFF. Calls are blocked until it is switched on."
5. The gate only applies when the company has recording switched on.

8.3 Alternative routes kept in the platform

Route	How it works	Status
SIM (default)	Phone dials from its SIM; phone recorder file uploaded.	In use.
Server callback (FreeSWITCH)	Server rings the employee's phone, then the customer, bridges and records centrally (both sides).	Built and tested against a simulator; needs a non-production FreeSWITCH to validate.
Cloud click-to-call (Exotel)	Exotel rings employee then customer; dual-channel recording copied to our storage.	Built; dropped by decision in Oct 2026.
Ametecs Click2Call / WebRTC	Own telephony service.	Considered; current decision is plain SIM calls.

9. Call outcomes, dispositions and follow-ups

Call result	Meaning
completed	Answered and ended normally.
no_answer	Customer did not answer (SIM route: talk time 0).
busy / rejected	Customer busy or rejected (server routes).
disconnected	Answered, then dropped abnormally.
agent_unanswered	Server routes only: the employee's own phone did not answer the callback; the customer was never dialled.
cancelled	Ended before the customer answered, or result never reported.
failed	Network or number failure.

Every call records **ring time**, **talk time** and **total time**. The phone's clock is not trusted beyond a 5-minute skew; the server corrects it.

After the call the employee must choose a **disposition**, can add notes, and must set a follow-up date when the disposition requires it. Default codes seeded for every company:

Code	Label	Follow-up required
CONNECTED_PTP	Promise to pay	Yes
CALLBACK	Call back later	Yes
NOT_REACHABLE	Not reachable	No
WRONG_NUMBER	Wrong number	No

Code	Label	Follow-up required
REFUSED	Refused to pay	No
PAID	Already paid	No

Companies can edit, add or deactivate codes. Outcomes can be entered offline and are sent when the phone reconnects.

10. Recordings

- Each call has a recording status: *not required, pending, recording, processing, available, failed, deleted by retention*.
- **A completed call never implies a successful recording.** If no file is found on the phone, the recording is marked Failed with the reason (never faked), and the console is alerted.
- Accepted phone formats: m4a, mp4, aac, mp3, wav, ogg/opus, amr/awb, 3gp — up to 50 MB. AMR/3GP play only after download (browsers cannot play them).
- Each file gets a SHA-256 fingerprint on arrival, so any later tampering is detectable.
- Playing or downloading uses a link valid for 5 minutes, and every play and download is audited. Play and download are separate permissions.
- Default retention is 180 days (policy). Deleting expired recordings needs a plan prepared by one admin and **approved by a different admin**. Recordings under legal hold are never deleted.

11. Supervision

11.1 Dashboard and status

The dashboard shows KPIs, a 7-day call chart and per-employee productivity, plus a green banner "N new messages from employees". **Status & location** shows each phone as observed: online/offline (offline after 3 minutes without contact; the app sends a heartbeat every 60 seconds), work state (signed out, idle, active, on call, updating customer), battery, network and last contact. "Signed in" does not mean "working" — the console says so.

11.2 Location

- Collected only during a work session, every 5 minutes by default, with a visible notification on the phone. Stops at sign-out.
- Background collection only if the policy allows and the permission is granted through device management — never by tricking the user.
- Viewing locations is audited. Default retention 90 days.

11.3 Messages and internet voice calls

- Supervisor ↔ employee chat with sent / delivered / read ticks and attachments (allowed types and size set by policy, re-checked on the server).
- Console: unread badge on the menu, chime, Windows desktop notification, unread count in the browser tab, mute toggle, "Mark all read".
- Phone: notification shows the actual message with **Reply** and **Mark read** buttons; tap opens the chat.
- Internet voice call (WebRTC) from the admin chat header to the employee's app: the phone rings with Answer / Decline, a missed-call notification appears if unanswered, and each call is logged in the chat ("Internet call · m:ss", "Missed internet call", "Call declined"). These calls are not recorded.

11.4 Screen and camera sessions

Only with consent, never hidden. The supervisor must state a purpose (min. 5 characters); the employee sees who is asking and why and can decline; Android then asks its own permission; a red banner and a notification with Stop stay visible. Unanswered requests expire after 90 seconds and are never run later. Screen sessions time out after 10 minutes (camera 5) by default; at most 3 screen sessions run at once per company. Screenshots are possible only inside an active screen session and only if policy allows.

12. Policies and compliance

A policy profile is a set of rules. The effective rule for a phone is worked out key by key: **phone override beats team override, team beats company default**. The console shows each value and where it came from.

Rule group	Examples (default)
Calling	Calling enabled (yes); allowed prefixes (+91); manual dial (no); calling hours.
Recording	Mode (mandatory); if it fails (continue and flag, or cut the call); notice text "This call may be recorded for quality and training purposes."
Location	During work session (yes); interval (300 s); background (no).
Supervision	Live screen (off), screenshots (off), camera sessions (off), timeouts, concurrent sessions.
Device (needs Android Enterprise)	App allowlist, block unknown apps, kiosk, camera/mic block, screenshot block, USB file transfer (blocked), SD card (blocked), Bluetooth config, location forced on.
Offline & revocation	Offline lease (240 min); active call on revoke (terminate).
Retention	Recordings 180 days; chat 365 days; location 90 days.

- Before publishing, the system blocks contradictions (e.g. camera blocked but camera sessions on; calling on with empty calling hours; employee app not in the app allowlist) and warns about risky ones, which must be accepted.
- States: Pending → Applying → Applied / Partially applied / Failed. **A phone is compliant only when every mandatory rule is applied and verified.** Until then the employee cannot see customers or call.
- Device-level rules on a phone without Android Enterprise enrolment are shown honestly as "Requires setup" — never as applied.
- Roll back republishes the previous version as a new, audited version.

13. Integration with SmartPRS2 and SmartEPT

System	Owns	Receives from Caller
SmartPRS2	Employee geofence (start point, radius, tolerance) and attendance of record for payroll.	Live sign-in geofence check; each login/logout as an IN/OUT punch (source "caller", with location and geo status).
SmartEPT	Enforcement and monitoring: violations, reports.	Each login/logout as a punch (like a biometric punch); every GPS fix to the field trail; leaving the fence raises a GEOFENCE_EXIT violation, coming back resolves it.
Caller	The phone, sign-in and location capture.	— (sender)

- Data is never lost when PRS or EPT is down: it waits in an outbox and is retried with growing gaps (1, 2, 4 … up to 60 minutes), indefinitely.
- Every event has a unique ID, so a retry never creates a second attendance row.
- Admins configure each connection under **Administration → Integrations (PRS / EPT)** with URL + API key, "Save & test", and "Send waiting data now".

> **Open decision.** If a company already pushes attendance from SmartEPT to SmartPRS2, field staff would be counted twice (sources "caller" and "smartept"). One path must be chosen per company.

14. Notifications

With Firebase Cloud Messaging switched on, the server wakes the phone through Google even when the app is closed, so messages and incoming internet calls ring immediately. The push carries **no customer data or message text** — only "something changed"; the phone fetches the details itself over a secure connection. Without Firebase the app still works by checking the server every 2–20 seconds while it is open.

15. Offline working

- The employee can keep working offline for the company's offline lease (default 4 hours). The app shows how old its data is and how many updates are waiting.
- Outcomes, notes, follow-ups, activity, location and read receipts are queued and sent later; each has a unique ID so nothing is counted twice.
- Anything recorded after the lease expired is rejected by the server — work done while unauthorised does not count.
- Calls need the server to authorise them, so a call cannot start offline.

16. Security, privacy and audit principles

- **No hidden surveillance.** No rooting, no secret camera or microphone, no auto-accepting Android consent prompts. The employee's *Monitoring* screen lists exactly what is enabled.
- **Company separation.** Every record carries its company; a request that cannot be tied to exactly one company is refused.
- **Strong device identity.** Knowing a phone's ID gives nothing; every sensitive request is signed by a key locked inside that phone.
- **Everything privileged is audited**: approvals, revokes, wipes, policy publishes, location views, recording plays/downloads, retention approvals.
- **No legal claims.** Recording notices are configurable text; the platform does not claim regulatory certification.

17. Admin console modules

Group	Module	Purpose
—	Dashboard	KPIs, 7-day chart, productivity, new-message panel
Team	Employees & teams	Users, roles, teams, supervisors
Team	Status & location	Presence, battery, location fixes, field trail
Team	Messages	Chat list + thread, internet voice call
Calling	Calls & recordings	Filters, inline player, durations, outcomes, retry fetch
Calling	Follow-ups	Due and overdue follow-ups
Calling	Configuration	Calling and recording settings, disposition codes
Customers	Customers & assignment	Search, assign, reassign, withdraw, customer page
Customers	Imports	Upload, map, validate, commit, error report
Devices	Devices & approvals	Review, approve, reject, reassign, revoke, wipe, effective policy
Devices	Policy profiles	Draft, conflicts, publish, assign, roll back

Group	Module	Purpose
Supervision	Live screen / Camera sessions	Consent-based sessions
Administration	Reports & exports, Branding, Storage & retention, Audit logs, Integrations (PRS/EPT), System health	Company administration

18. Deployment and commercial model

Model	Description	Fits
Cloud SaaS (shared)	Many companies on one Ametecs server; each has a workspace address (company.domain) or verified custom domain.	Small and mid-size agencies; per-user monthly subscription.
On-premises (fixed)	One company on its own server; recordings on its own disk or storage.	Banks, large agencies with data-residency needs; perpetual licence + AMC.

The product name and logo are configurable, so the platform can be sold under Ametecs branding or white-labelled. It strengthens the Ametecs ecosystem story: SmartPRS for payroll and attendance, SmartEPT for enforcement, SmartDCM for the collection CRM, and this platform as the controlled mobile calling and field-force layer.

> **Important.** Full device lock-down (blocking uninstall, permission changes, factory reset) requires Android Enterprise "fully managed" enrolment via the Android Management API, which Google allows only for commercial EMM offerings sold to customers, with a business justification for the first 500 devices. Ametecs' eligibility must be confirmed before promising it to clients.

19. Current status and open business decisions

19.1 Status as of 7 October 2026

Area	Status
Server, admin console, Android app foundation	Built; automated tests passing; live end-to-end run on Laragon (1 Oct).
SIM calling with phone recorder + strict recording gate	Built; on-device testing on OPPO, OnePlus and Samsung pending latest build.
SmartPRS2 / SmartEPT field-force integration	Built and delivering in development; PRS test of the new hold-and-map case and roll-out to live PRS clients pending.
Messages, notifications, internet voice call	Built; admin rebuild and Android compile/test pending.
Firebase push	Code ready; Firebase project setup pending.
Android Enterprise lock-down (QR enrolment)	Designed; not started.
21 manual integration test cases	Not run yet.

19.2 Decisions needed

1. Choose one attendance path for field staff (Caller → PRS, or EPT → PRS) to avoid double attendance.
2. Keep fail-open sign-in when SmartPRS2 is down, or offer a strict per-company mode.
3. TURN server for internet calls over mobile data (today they work reliably on the same Wi-Fi).
4. Android Enterprise enrolment (QR) to block uninstall and permission changes — and confirm Google eligibility.
5. Production hosting (move from the development server to a proper web server; cloud region; storage).
6. Final product name and logo.